SE301 ADVANCED PROGRAMMING AND DESIGN · PROJECT 2026-27

Singapore Vertical Farming Research Management System

Legacy Refactoring & Modernisation Report

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1. Legacy System Analysis

The system keeps each growing chamber of a vertical farm within its crop's required conditions: it monitors sensors against crop profiles, runs devices to correct deviations, raises alerts and reports on resource use. The legacy SVFRMS simulates one day of this. Four chamber threads (CH1 Tomato, CH2 Strawberry, CH3 Lettuce, CH4 Chili) each run 20 cycles. Each cycle reads five sensors (temperature, humidity, soil moisture, light, nutrient), checks every reading against hard-coded crop thresholds, raises alerts, charges water and energy, and puts device requests (for example `"CH1|Cooling Fan"`) on a bounded queue of 5. A control thread takes requests off the queue and runs the device, which changes the chamber's state. At the end, two racing threads archive the alerts and print the daily report, and `archive/daily_report.txt` is written.

Everything is in one 393-line class, `SVFRMS.java`. Its main components are global `public static` state (lines 7–31), the `ChamberState` / `ChamberSummary` model (33–46), a string-switched event hub `handleEvent` (62–88), the rule engine `processSensor` (141–205), device control `activateControl` (90–129), a hand-written queue and `ControlThread` (131–139, 251–273), `ChamberThread` (275–295), reporting (207–241, 309–347), dead code (243–249, 297–307), and `main` (349–392).

2. Code Analysis and Limitations

2.1 Design problems (SOLID, spec 4.2)

Problem	Evidence in legacy code	Principle
God class	One class holds the model, rules, devices, queue, threads, reports and file I/O. Any change means retesting everything.	SRP (i)
Crop `if/else` chains	Adding a crop means editing `processSensor` in up to 3 places (147–187). The same alert block is copy-pasted 19 times, with thresholds and costs as magic numbers.	OCP (ii), DRY
Device `if/else` chain	`activateControl` (90–129) must be edited for every new device; each branch repeats the same three `handleEvent` calls.	OCP (ii), SRP
`extends Thread`	`ControlThread` swallows `interrupt()` and loops `while(true)`, breaking the `Thread` postcondition that `interrupt()` cancels it, so it cannot stand in where a cancellable `Thread` is expected (251, 258).	LSP (iii)
One wide entry point	`handleEvent(type, chamber, crop, msg)` does logging, storage, counting and printing chosen by a string; callers pass dummy `""` (95, 142, 148).	ISP (iv), SRP
Global mutable state	Static lists, maps, counters, `System.out`, `new Date()`, a static `Random`: nothing can be tested or replaced.	DIP (v)
Stringly-typed messages	Queue items are `chamber + "|" + device`, split again by the consumer (136, 269).	–
Duplicated report, dead code	Report format written twice (210–214 vs 314–319); four unused methods and two unused locks.	SRP

2.2 Concurrency problems (spec 4.4)

We ran the unmodified program 300 times and checked each run against invariants derived from its own archived alert list. **Only 92 of 300 runs were correct.**

#	Issue	Effect	Observed
C1	Check-then-act on `HashMap` (`containsKey → put → get`, 48–60) from 4 threads	`get` returns `null`; the chamber thread dies with an NPE (146, 80) and the chamber vanishes from the report	123/300 runs
C2	Unsynchronized `ArrayList.add` (64, 73)	Lost alerts: generated count ≠ archived count	67/300

C3	Non-atomic `total += n` counters	Lost water/energy updates	27 / 24 of 300
C4, C5	`ChamberState` (C4) and `ChamberSummary` (C5, 79–86) written by chamber and control threads without a lock	`temperature - 5` uses a stale value or overwrites a newer reading; summary counts wrong via C1	C4: seeded runs differ; C5: via C1
C6	Archive and report threads started together (384–386); a lock gives exclusion, not order	Report sometimes printed before archiving	7/300 (19/500 overall)
C7, C8	`alertLock` taken by readers but never by writers (C7); `waterPumpLock`, `nutrientLock` never used (C8)	False sense of safety	inspection
C9	`InterruptedException` swallowed in both `wait()` loops	Threads cannot be cancelled	inspection
C10	**Deadlock:** if the control thread dies (e.g. C1's NPE in `activateControl`), producers `wait()` forever on the full queue and `main` forever in `join()`	Program never ends, no report	reachable via C1
C11	No failure handling for threads	A crashed chamber still produces a "successful" report	every C1 run

Deadlocks. *Reachable:* C10. Producers wait for a slot only the dead consumer could free and `main` waits for the producers; nothing times out, and C9 means it cannot be cancelled. *Latent:* `alertLock → reportLock` is taken in the same order in both report methods, so it cannot cycle as written, but reversing either would let the two report threads each hold one lock and wait for the other (the commented-out `sleep`s at 224 and 385 widen that window). Section 5 removes both.

Spec gaps. Missing: severity levels and the sensor-failure alert (3.4); Heating Unit, Ventilation System and Shade Controller (3.3); all 3.5 reports but the daily one. We add them all without touching the legacy output (section 6); the Weekly Crop report sums the last seven simulated days (one per run) rather than inventing data. Kept as-is because spec 4 requires preserved behaviour: only Tomato has every rule, and a stale Cooling Fan or Grow Light request can move a newer reading the wrong way (the new devices guard against this). **Deliberate changes:** failure paths that hid defects now exit with status 1: a failing alert listener or device (legacy: silently killed its chamber, C11, or the control thread, C10) and an unsaved report (legacy: exit 0).

3. Proposed Design

Design rule: preserve behaviour exactly, and fix every smell and concurrency defect above. Every class exists to remove a specific problem. We captured the legacy output with a fixed seed and serial schedule, and our refactored output is **byte-identical after normalization** (timestamps and control-thread timing removed).

- **Crop rules as data.** All 19 copied alert blocks became rows in `crops.csv`: `crop, sensor, HIGH|LOW, limit, device, water, energy, message`. `*` rows apply to every crop (the legacy global light and nutrient limits). `SensorProcessor` loops over whatever rules the registry holds and never names a crop; each rule's `Device` is resolved once at load time. We chose a CSV over a Java catalogue class because adding a crop to a class is still "modifying existing code" (spec 4.2 ii).
- **One class per device** behind a small `Device` interface (`name`, `apply`, and the `effects` it declares, e.g. *lowers TEMPERATURE*). `ControlService` only queues and runs requests; what happens after a device runs (summary, console lines) is up to its `ActionListener`s.
- **Observer for alerts**: one `Alert`, and each listener does one job. **Objects, not strings, on the queue**: a `ControlRequest`.
- **Composition root.** Only `SVFRMS.createSimulation` creates and connects objects; everything else gets its dependencies through its constructor, so the tests and the baseline harness can inject a seeded or fixed sensor

source.

- **Fail fast on bad configuration.** Every CSV error (unknown device, a device that cannot correct its rule's breach, wrong columns, bad number, duplicate rule or chamber) stops the program before any thread starts, naming the file and line.

3.1 SOLID mapping

Spec	Change
(i) SRP	56 small classes and interfaces in 7 packages plus the `SVFRMS` entry point, each with one job. All report formatting lives in `DailyReport` only; `ControlService` only queues and runs requests.
(ii) OCP	New crop = rows in `crops.csv`. New device = new `Device` class + one line in its `ServiceLoader` file. New alert output = new `AlertListener`. No existing class changes.
(iii) LSP	No `Thread` subclasses: tasks are `Runnable`s run by an `ExecutorService`, and the consumer exits on interrupt. `Device` has only methods every device can honour (no `increase()/decrease()` pair a fan cannot implement); a test checks that each device does exactly what its `effects()` declare.
(iv) ISP	`handleEvent` is replaced by small 1–4 method interfaces: `SensorSource`, `Device`, `AlertListener`, `ActionListener`, `ControlSink`, `UsageMeter`, `ReadingLog`, `UsageTotals`, `AlertArchive`, `SummarySource`, `CropRules`.
(v) DIP	The rule engine sees only abstractions: `SensorProcessor` reads `CropRules`, sends alerts to an `AlertListener`, charges a `UsageMeter`, queues on a `ControlSink` and logs readings to a `ReadingLog`; `ControlService` runs any `Device` and notifies `ActionListener`s; reports read `AlertArchive`, `SummarySource` and `UsageTotals`; the nutrient report finds its devices by effect, not class. `device` and `sensor` never depend on `report`; `Clock`, streams and sensors are injected.

4. Final Architecture

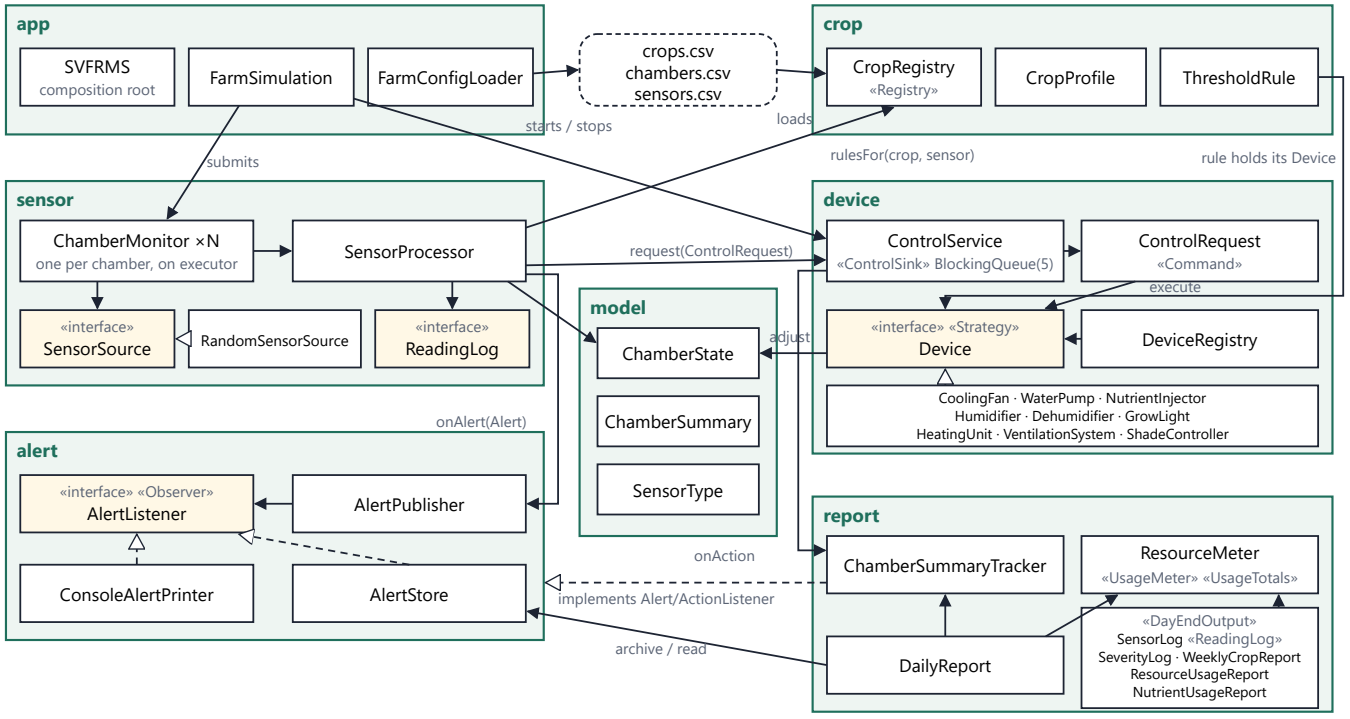


Figure 1. Component diagram. Yellow boxes are interfaces; dashed arrows with hollow heads mean "implements".

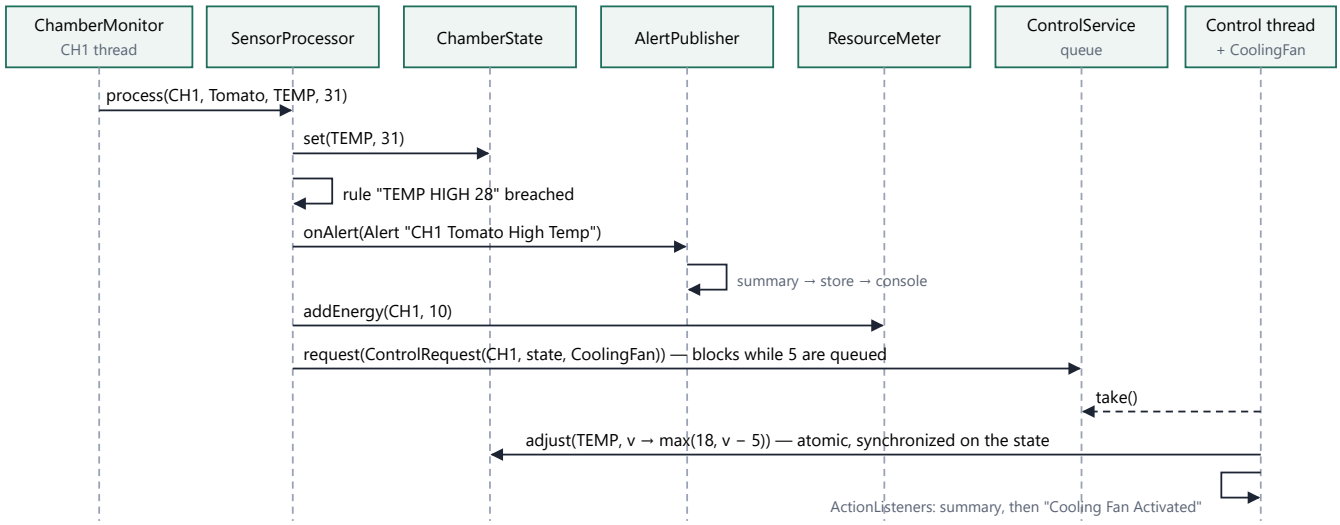


Figure 2. Sequence: one high-temperature reading in CH1. No lock is held while publishing or queueing.

End of the run (`FarmSimulation.run`). The control consumer starts and one `ChamberMonitor` is submitted per row of `chambers.csv`; the main thread waits on each `Future`. `control.stop()` then queues a stop marker *behind* all pending requests, so the queue drains first (it gives up after 30 s if a device hangs), and the control thread's own `Future` is checked. Finally the main thread, in a fixed order, archives the alerts, prints the report and saves it, plus `alerts_by_severity.txt`, `sensor_log.txt`, `resource_usage.txt`, `nutrient_usage.txt` and `weekly_crop_report.txt`. **Exit status**: 1 if a chamber, the control thread, a device, an alert or action listener, or a report save failed (the error is printed); 2 for an invalid CSV file; 3 for an internal error (a built-in resource missing or unreadable, or a bad or duplicate device). A chamber that stopped before its 20th cycle without failing is named as a partial day.

5. Concurrency Fixes

Issue	Fix	Tool
C1	Summaries created with atomic `computeIfAbsent`. Chamber state is never looked up: each monitor owns its `ChamberState` and passes it inside the `ControlRequest`.	`ConcurrentHashMap`
C2, C7	All alert-store methods synchronized on one lock that guards active list, archive and count together; the sensor log is synchronized the same way (timestamp formatted outside the lock).	intrinsic lock
C3	Water and energy, one pair of counters per chamber created atomically; the totals are their sum, so both reports read the same counters.	`AtomicInteger.addAndGet`, `computeIfAbsent`
C4, C5	`ChamberState.set/adjust` and `ChamberSummary` methods are synchronized (reports read one `snapshot()`); a device's read-modify-write is one call to `adjust` (the Ventilation System changes both readings in one `changeAll` call, which runs no outside code under the lock).	one lock per chamber
C6	Archive, report and save run sequentially on the main thread after all workers finish.	program order
C8–C10	Unused locks removed and the hand-written buffer replaced. The consumer catches a failing request, counts it (the run then fails) and keeps going, and executors shut down with a timeout. Interrupts are never swallowed: a producer interrupted on a full queue keeps the flag and fails loudly, so no request is lost silently. If the consumer dies, a waiting producer stops and fails instead of blocking forever (the C10 deadlock), and every request still queued is reported and fails the run. If the stop marker cannot be queued within 30 s, the consumer is interrupted instead. An interrupted run stops the chambers before the consumer and still writes its report: each chamber checks the flag before every reading, the report waits for the chamber threads to stop, and any that stopped early is named.	`ArrayBlockingQueue`, `ExecutorService`

C11	Each chamber's `Future` holds its exception; once every thread has stopped, each one is reported (even if the run was interrupted first) and the exit status is non-zero.	`Future`

Deadlock freedom by construction. No method holds two locks at once. Each lock (`ChamberState`, `ChamberSummary`, `AlertStore`, the sensor log) is held only for a few field updates and never while calling another object; `SensorProcessor` holds no lock while it publishes an alert or waits to queue a request. With at most one lock held at a time, the lock-wait graph cannot contain a cycle.

> **Evidence.** Legacy: 92/300 concurrent runs correct. Refactored `run.jar`: **300/300** correct with the 4 legacy chambers and **300/300** with the 7 demo chambers, with the same invariant checker. Seeded serial runs: normalized output **identical** to the legacy baseline in **42/42** comparisons (21 seeds × the 4 legacy chambers and all six legacy crops); the checker prices alerts from the legacy source, not our `crops.csv`. *Limits:* device lines are compared as counts (their values depend on thread timing), the concurrent run only against invariants (legacy output varies run to run), and archive-before-report is the legacy race's usual order (~96%), now fixed. 89 JUnit test methods (101 runs, as the stress tests repeat 3 times), including stress tests that hammer every shared object from 8 threads × 5,000 operations, a producer-consumer test with a queue of 1, a failing-device test (the run fails) and a crashed-consumer test (C10), an interrupted-producer test (C9), a failing-chamber test (C11, also when the run is interrupted), an interrupted-run and two interrupted-chamber tests (between readings, and while queueing a device request), and a failing-alert-listener test that still fails the run.

6. Extensibility and Maintainability

Change	Legacy	Refactored
Add a crop (Kale)	Edit `processSensor` in up to 3 places	Add rows to `crops.csv`, and a row such as `CH5, Kale` to a `chambers.csv`
Remove a crop	Delete branches and hope nothing else uses them	Delete its rows
Change a threshold or cost	Find the right `if` among 19 copies	Edit one number in one row
Add a device (Heating Unit)	Edit `activateControl`	New `HeatingUnit implements Device`, list it in `META-INF/services`, use its name in a CSV row; no existing class changes (done for Heating Unit, Ventilation System, Shade Controller)
Add a report (e.g. weekly)	Edit `main` and the report code	New class reading `UsageTotals` / `SummarySource` (or an `ActionListener`), implementing `DayEndOutput`; register it in `SVFRMS` (done for water, energy, nutrients and the weekly crop report)
Add an alert output (file, SMS)	Edit `handleEvent`	New `AlertListener`, subscribe it in `SVFRMS` (done for severity, below)
Use real sensors	Rewrite `ChamberThread`	Implement `SensorSource`, pass it to `createSimulation`
Change the report format	Edit two places	Edit `DailyReport`

Crop extension (spec 4.5). Kale, Bok Choy, Bell Pepper and Eggplant were added as rows in `crops.csv`, with **no Java changes**; each automatically gets the shared light and nutrient rules. The default `chambers.csv` keeps the 4 legacy chambers, so a plain run behaves exactly as before. `java -jar run.jar demo` reads the `demo/chambers.csv` built into the jar, which adds `CH5, Kale`, `CH6, Bok Choy` and `CH7, Bell Pepper` without a rebuild (a notice names the override file); this 7-chamber layout also passes 300/300 concurrent runs. Only an explicitly named folder is read, so a stray CSV can never change a normal run. A chamber whose crop has no rows (a removed crop or a typo) keeps only the shared rules, as in the legacy code, and a start-up warning names the line. Bell Pepper and Eggplant use the three new devices (Heating Unit for low temperature, Ventilation System for high humidity, Shade Controller for high light); since a request runs after the chamber may have stored a newer reading, a device never moves a

reading the wrong way (a heater never cools). A row whose device would push the reading further out (e.g. a Cooling Fan on a LOW rule) is rejected at start-up.

```
Kale, SOIL, LOW, 40, Water Pump, 30, 0, WARNING, {crop} Low Soil Moisture → [CH5][Kale][ALERT] CH5 Kale Low Soil Moisture
```

Alerting extension (spec 3.4). Each rule now carries a severity column (temperature CRITICAL, humidity, soil and nutrient WARNING, light INFO). A new `SeverityLog` listener writes every alert with its level and a count per level to `archive/alerts_by_severity.txt`. A sensor that does not respond, or reports a reading outside its valid range in `sensors.csv` (NaN, humidity outside 0–100 %), raises a CRITICAL *Sensor Failure* alert; nothing is stored, no device runs and the chamber carries on. `demo/sensors.csv` models a light sensor that reads at most 1100 lux, so the demo shows these alerts. Each extension file adds one console line after the legacy output (five in all), e.g. `Alerts by severity: INFO=21, WARNING=64, CRITICAL=93`; the sensor log the legacy code collected but never used is saved to `archive/sensor_log.txt`; every legacy console line and the daily report are still identical to the legacy baseline.

Reporting extension (spec 3.5). Three new `DayEndOutput`s, registered in `SVFRMS`, with no change to `FarmSimulation`: `ResourceUsageReport` (Water Usage and Energy Consumption per chamber, `resource_usage.txt`; this needed one change underneath: `UsageMeter`'s methods now take the chamber id), `NutrientUsageReport` (also an `ActionListener` counting runs of every device that raises NUTRIENT, `nutrient_usage.txt`) and `WeeklyCropReport` (alerts, actions, water and energy per crop over the last seven complete runs, one simulated day each, kept in `weekly_crop_history.csv`; plus each crop's spec 3.1 profile, e.g. *Temperature 22–28 C, Soil ≥ 45 %*, derived from its rules).

7. Design Patterns

Pattern	Where	Problem it solves and why we chose it
Strategy	`Device` + 9 classes	The `activateControl` chain. Each device is a different algorithm behind one interface, so a new device never touches `ControlService`. Devices are found with `ServiceLoader`, so adding one edits no existing class.
Observer	`AlertPublisher` → `AlertListener`s	`handleEvent` did four jobs behind a string switch. The rule engine now raises one alert, and each listener (summary, store, console) does one job. New outputs are new listeners; a failing listener is reported on its own, the rest still run, and the run fails. Device actions use the same pattern (`ActionListener`: summary, console, nutrient report).
Command	`ControlRequest`	Replaces the `"CH1|Cooling Fan"` string. The request carries the device and the chamber state and has `execute()`, so the producer decides *what* and the consumer decides *when*.
Producer–Consumer	`ControlService`	Replaces the hand-written `wait/notifyAll` buffer that swallowed interrupts, with the JDK's `ArrayBlockingQueue` (capacity 5, as before).
Registry	`CropRegistry`, `DeviceRegistry`	Crop and device names were hard-coded in control flow. A lookup by name is all that is needed; no factory hierarchy. `DeviceRegistry.discover()` fills itself from `META-INF/services`; a duplicate device name is rejected at start-up.
Composition root (DI)	`SVFRMS.createSimulation`	Replaces global static state. One place builds the object graph, so tests and the baseline harness can swap the sensor source.

Not used on purpose: a Strategy class per crop (crops differ only in numbers, so data is simpler); Singleton (it would recreate the global state we removed); Factory (nothing needs a family of products).

Build and run: `mvn package` (runs the tests and builds `target/run.jar`), then `java -jar run.jar` (or `java -jar run.jar demo` for the new crops). Evidence: `baseline/`; full analysis: `docs/phase1-analysis.md` and `docs/phase2-design.md`.